



Cambrian College

School of Business and Information Technology, Creative Arts, Design, Music and Hospitality

Course Outline

Cambrian College would like to acknowledge that we are situated on the traditional lands of the Robinson Huron Treaty, and acknowledge our host, Atikameksheng Anishnawbek, as well as our ancestors of this land, and the Anishnaabe people.

Course Title	Dashboards and Data Analysis				
Course Code:	EXL1002	Credit Value:	3	Credit Hours:	42
Programs:	BAPG Business Analytics CAGC Crime Analytics HAGC Health Analytics				
Equivalencies		Prerequisites		Corequisites	

This course may be delivered in a variety of different formats: 100% in-class, 100% online (or a blend of both), videoconferencing, distributed learning or off-campus. Please confirm with your faculty member which format will be used for your section of this course.

General Education Course: ☐

Degree Breadth Course: ☐

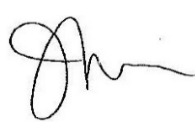
Eligible for PLAR: ☒

COURSE DESCRIPTION

In this course, students will build on Excel fundamentals and learn how to create meaningful spreadsheets, dashboards, and visualizations to collect, analyze, manage and communicate data. Students will learn how to organize and consolidate data in multiple worksheets; perform visual analysis; apply logical operators and formulas; work with text and financial formulas; work in an object-linking and embedding environment; apply pivot tables to data; create dashboards and data summaries, utilize goal seek and what-if analysis.

Date: May 27, 2021

Prepared By: Sidney Shapiro

Approved by: 

JILL FERGUSON
Chair, School Of Business And Information Technology,
Creative Arts, Design, Music And Hospitality

Effective: Fall 2021, Winter 2022, Spring 2022

RELATIONSHIP TO PROGRAM VOCATIONAL LEARNING OUTCOMES

PROGRAM LEVEL	
This course contributes to your program by allowing you to demonstrate the following vocational learning outcomes:	
Program(s)	Vocational Learning Outcomes
Business Analytics	College Standards 1. Collect, manipulate and mine data sets to meet organizational need. 2. Recommend different systems architectures and data storage technologies to support data analytics. 3. Design and apply data models that meet the needs of a specific operational/ business process. 4. Develop software applications to manipulate data sets, correlate information and produce reports. 5. Design and present data visualizations to communicate information to business stakeholders. 6. Apply data analytics, business intelligence tools and research to support evidence-based decision making. 7. Identify and assess data analytics business strategies and workflows to respond to new opportunities or provide project solutions.
Crime Analytics	College Standards 1. Collect and analyze a variety of data sets to identify information related to crime analysis. 2. Design and apply data models that meet the needs of a specific operational process. 3. Design and present data visualizations to communicate information to stakeholders. 4. Create effective oral presentations and produce fitting analytical products suitable for specific audiences.
Health Analytics	College Standards 1. Collect and analyze a variety of data sets to identify information related to organizational health needs. 2. Mine and manipulate data sets, correlate information and produce outputs using advanced software functions. 3. Design visualizations and communicate results of data analysis to decision makers, stakeholders and the public. 4. Apply health analytics and business intelligence tools to support evidence-informed practice.

COURSE CURRICULUM

Topics/Concepts Covered in This Course

- Synthesize data

Topics/Concepts Covered in This Course

- Identify patterns and manipulate sets of data
- Advanced formulas
- Macros
- Develop dashboards using various scenarios
- Tables and pivot charts/tables
- What-if analysis, goal seek, data tables
- Visual data analysis using Power View
- Connecting to and using external data

COURSE LEVEL: Learning Outcomes and Objectives	
To earn credit for this course, you must reliably demonstrate your ability to:	
Learning Outcome	Objectives
1. Analyze, summarize and understand complex data	1.1 Synthesize data 1.2 Simplify and break down complex data 1.3 Describe data examples from different industries and in different contexts 1.4 Utilize visual design tools
2. Work with different types of data	2.1 Import various types of data 2.2 Explain what steps to take to perform an analysis 2.3 Discuss data mining 2.4 Explain the use of different models for various data sets 2.5 Identify where to gather data 2.6 Discuss how various types of data can be combined 2.7 Discern patterns in data 2.8 Identifying proportions in different sets of data and within data sets using references 2.9 Identify appropriate methods to present data relationships 2.10 Explain differences in data and demonstrate ways to illustrate differences
3. Describe and demonstrate select advanced features of Microsoft Excel	3.1 Demonstrate a familiarity with Microsoft Excel basic program functions 3.2 Apply range and formula functions as they pertain to data analysis and Excel functionality 3.3 Identify appropriate formulas for data specific problems 3.4 Apply Excel formulas for basic operations, statistics, math, text, date and time, and conditional analysis 3.5 Create various lookup formulas 3.6 Develop pivot tables and pivot charts 3.7 Discuss basic data arrays, data cubes, and high dimensional data

Learning Outcome	Objectives
	3.8 Create and record macros
4. Visually analyze and present data	4.1 Create pivot tables and pivot charts visual based on provided data 4.2 Create graphs in Excel using provided data 4.3 Plan a data analysis using the best approach 4.4 Develop clear and concise data reports using multiple data sources
5. Develop data summary dashboards	5.1 Determine what information is required in the construction of a dashboard 5.2 Analyze data options to consider key performance indicators to present it in a dashboard 5.3 Demonstrate the tools required to create a dashboard in Excel 5.4 Apply various chart and graphical display options in Excel to display summary data 5.5 Assemble summary dashboards in Excel using various scenarios

Essential Employability Skills

Communication

- communicate clearly in written, spoken, and visual form that fulfills purpose/needs of audience.
- respond to written, spoken, or visual messages in a manner that ensures effective communication.

Numeracy

- execute mathematical operations accurately.

Critical Thinking and Problem Solving

- apply a systematic approach to solve problems.
- use a variety of thinking skills to anticipate and solve problems.

Information Management

- locate, select, organize, and document information using appropriate technology and info systems.
- analyze, evaluate, and apply relevant information from a variety of sources.

Interpersonal

- show respect for the diverse opinions, values, belief systems, and contributions of others.
- interact with others in groups in ways that contribute to effective working relationships.

Personal

- manage the use of time and other resources to complete projects.
- take responsibility for one's own actions, decisions, and consequences.

Delivery Method

- Classroom: Course is delivered through scheduled synchronous teaching that may be face-to-face and/or virtual.
- Online: Course is fully delivered through asynchronous teaching.
- Hybrid: Course combines scheduled synchronous and unscheduled asynchronous teaching.
- HyFlex: Course includes both synchronous and asynchronous learning and the student can move between both components seamlessly.

Learning Activities

- Lectures
- Class Discussions
- Labs
- Group Work
- Research
- Self-Directed Learning
- Presentations
- In-Class Exercises
- Case Studies
- eLearning Components

Resources Required

Books

Joseph M. Manzo, *Data Analytics Using Microsoft® Excel®: With Accounting and Finance Datasets v2.0*, Published: 2019, Flatworld
ISBN: 978-1-4533-9522-6

Evaluation Plan

Grading Scheme

A	80% - 100%
B	70% - 79%
C	60% - 69%
D	50% - 59%
F	0% - 49%

Evaluation Method	Value (%)
Assignments	60%
There will be 6-10 assignments in this course. Please see your instructor for details as the assignments may be in the form of dashboards, written case studies, presentations, individual or group projects, or another form not listed.	
Test/Exam	40%
There will be multiple quizzes/tests/exams or another form not list and activities in this course. Please see your instructor for details.	

ADDITIONAL INFORMATION

College

Modifications to the Course Evaluation Plan

Under exceptional circumstances, Cambrian reserves the right to alter the Course Evaluation Plan. Upon approval from the Dean, faculty members will notify students of any modification to the Course Evaluation Plan and post to the course Learning Management System site (Moodle).

Academic Policies

Students must adhere to all academic policies. These are available on myCambrian as well as the College website at <https://cambriancollege.ca/about/official-documents-and-policies/academic-policies/>.

Accessibility Services

Cambrian is committed to creating an inclusive learning environment where we support our students' success. Students who are registered with the Glen Crombie Centre can access accessibility and/or counselling services by:

1. Registering and scheduling an appointment through Clockwork in the Student Portal in myCambrian.
2. Contacting 705-566-8101 ext. 7420 or 7311, or
3. E-mailing accessibilityservices@cambriancollege.ca or Counselling@cambriancollege.ca

Attendance

Regular attendance is strongly encouraged as it contributes to student success. Some courses have specific lab/participation requirements. Students are expected to be aware of, and adhere, to these requirements.

Audio/Visual Capture

Sounds and images from this class, and contributions made by a participant, virtually or in-person, are recorded under the authority of the Ontario Colleges of Applied Arts and Technology Act, 2002. The main purpose for these recordings is to allow students enrolled in this course, whether they attend the particular class in person, virtually, or at all, to review class content and activity. The use of class recordings is for personal use only and shall not be shared or transferred.

These recordings may also be reviewed by faculty members to prepare future classes, to evaluate students, to collaborate in program/course development/review, or to provide feedback to students or other faculty. Any questions about this collection may be addressed to your respective Dean.

Copyright

Students may not submit work or partake in activities that infringe on the Canadian Copyright Act, or the related Cambrian College Fair Dealing Guidelines and Copying Guidelines. The guidelines may be viewed at: <http://cambriancollege.libguides.com/copyrightinfo> or by contacting the Library for questions related to copyright: library@cambriancollege.ca

Wabnode Centre for Indigenous Services

Students are encouraged to come to Wabnode to meet our Elders and to access personal counselling and connect with academic tutoring. Our team is also able to provide students with information related to funding and scholarship and bursary opportunities for Indigenous students. We encourage students to come to Wabnode and find out about opportunities to share in cultural, sports and community activities. Students can also contact Wabnode at wabnode@cambriancollege.ca

Out-of-Class Assistance

The faculty will inform students of their Out-of-Class availability through the Learning Management System site (Moodle).

Prior Learning Assessment and Recognition (PLAR)

Students wishing to have work or life experience considered through Prior Learning Assessment and Recognition should contact pathways@cambriancollege.ca

Testing Practices at Cambrian

Many courses include major tests and/or final exams. The practice at Cambrian requires that these types of test situations involve proctoring to ensure academic integrity. Online tests/exams may employ a proctoring service to enable you to take your exam from a location of your choosing within a period specified by your instructor. When you are taking an online test/exam, the proctoring service may capture your video, screen, audio and web surfing particulars to protect academic integrity. Cambrian College collects, uses, discloses and retains personal information in compliance with the Freedom of Information and Protection of Privacy Act (FIPPA). Your personal information is being collected under the authority of the Ontario Colleges of Applied Arts and Technology Act S.O. 2002, c.8, Sched. F. This information will be used for the purpose of administering a test/exam through an online proctoring service acting as an authorized agent of the College. Please refer to Cambrian's Confidentiality of Student Records Policy for more details. If you have any questions regarding the collection of your personal information, please contact Vice President Academic, Cambrian College, 1400 Barrydowne Rd., Sudbury ON P3A 3V8, 1-705-566-8101.

Tests and Evaluations

Students are required to write tests and/or complete evaluations as scheduled. Exceptions may be made in the event of an emergency or a sanctioned event (e.g., varsity sports, field trip, religious observances, etc.). In non-emergency situations, students are expected to contact the faculty member in the event that they can not be present for a test/evaluation to explain the reason for their absence. At the discretion of the faculty member, an alternate time for the test/evaluation may be allowed.

Transfer Credit

Students wishing to have courses from other programs or institutions assessed for equivalency and/or transfer credit should contact transfercredits@cambriancollege.ca.

Equity, Diversity and Inclusivity

Cambrian is committed to building and preserving an equitable, diverse, and inclusive learning community where students, faculty, and staff may achieve their full potential in an environment characterized by equality of respect and opportunity. All students and employees have the right to live and work in an environment that is free from discrimination and harassment. Therefore, Cambrian College will not tolerate any form of discrimination or harassment in its employment, education, accommodation, or business dealings. For more information, please visit: <https://cambriancollege.ca/about/official-documents-and-policies/equity-human-rights-and-accessibility/>